

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study
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1 Objective

- Implement a Decision Tree classifier on the Wisconsin Breast Cancer dataset.
- Extend the analysis to a Random Forest ensemble model.
- Study how hyperparameters influence overfitting and generalization.
- Select the best hyperparameters using 5-Fold Cross-Validation.
- Compare the two models on standard evaluation metrics.

2 Problem Statement

Breast cancer diagnosis based on cell nucleus measurements is a standard binary classification task. Given 30 numerical features computed from a digitized image of a fine needle aspirate (FNA), the goal is to classify each sample as either **Benign (B)** or **Malignant (M)**. The experiment trains two tree-based classifiers, tunes their hyperparameters via cross-validation, and evaluates which one generalizes better on held-out data.

3 Dataset Description

Attribute	Detail
Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository
Number of Samples	569
Number of Features	30 (numerical, real-valued)
Number of Classes	2 — Benign (357), Malignant (212)
Missing Values	None
Train-Test Split	80% Training (455 samples) / 20% Testing (114 samples)

The features describe properties such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension — each computed as mean, standard error, and worst value for each cell nucleus.

4 Exploratory Data Analysis

4.1 Class Distribution

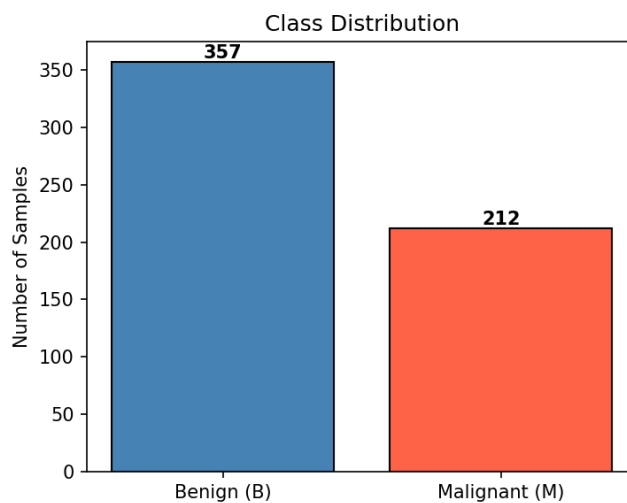


Figure 1: Class distribution of the WDBC dataset.

Inference: The dataset contains 357 Benign and 212 Malignant samples, giving a roughly 63:37 split. The class imbalance is moderate — not severe enough to require resampling, but it does mean a naive all-Benign classifier would already achieve 62.7% accuracy. This motivates using F1-score alongside accuracy for a fairer comparison. Stratified splitting ensures both train and test sets preserve this ratio.

4.2 Feature Correlation Matrix

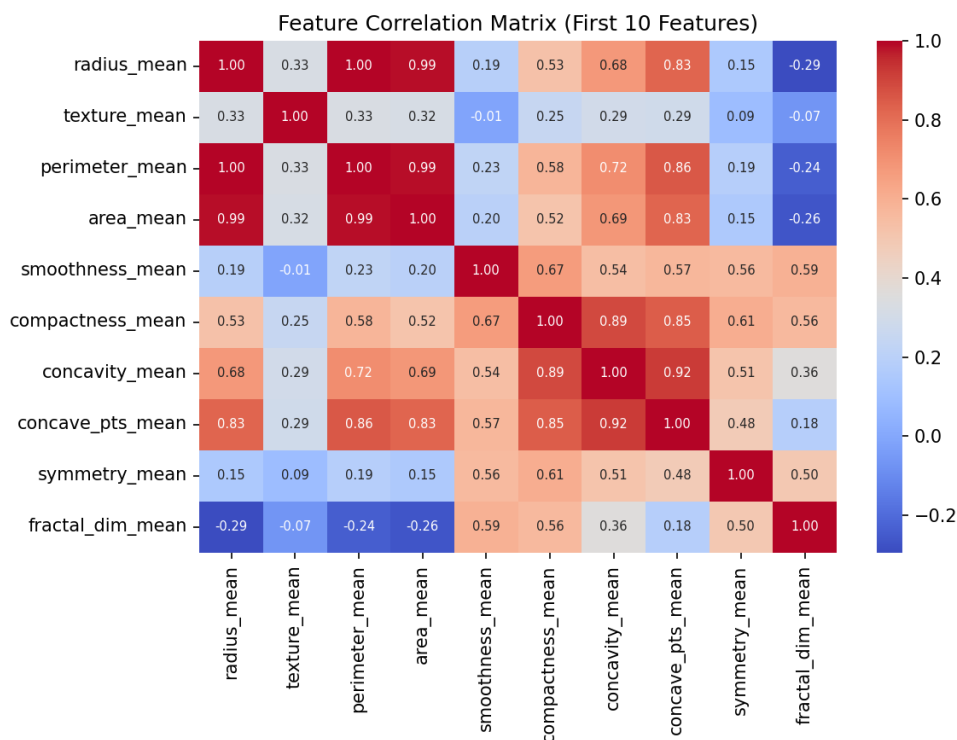


Figure 2: Correlation heatmap for the first 10 mean features.

Inference: Several features show strong positive correlation — for example, radius, perimeter, and area are near-perfectly correlated ($r > 0.99$) because perimeter and area are geometrically derived from radius. Compactness and concavity also correlate strongly. This multicollinearity does not hurt tree-based models (which split one feature at a time), but it does confirm that not all 30 features carry independent information. Random Forest’s random feature sub-selection at each split naturally handles redundant features.

5 Data Preprocessing

- **Label Encoding:** The target column `diagnosis` holds string labels M and B. These were encoded to integers using `LabelEncoder` — $B \rightarrow 0$, $M \rightarrow 1$.
- **Missing Values:** No missing values were present; no imputation was needed.
- **Feature Scaling:** Not applied. Decision Trees and Random Forests are scale-invariant since they rely on threshold comparisons rather than distance metrics.
- **Train-Test Split:** The dataset was split 80-20 using `train_test_split` with `stratify=y` and `random_state=42` to ensure reproducibility and balanced class proportions.

```
1 col_names = ['id', 'diagnosis'] + [f'feature_{i}' for i in range(1, 31)]
```

```

2 df = pd.read_csv('wdbc.data', header=None, names=col_names)
3 df.drop(columns=['id'], inplace=True)
4
5 le = LabelEncoder()
6 df['diagnosis'] = le.fit_transform(df['diagnosis']) # B=0, M=1
7
8 X = df.drop(columns=['diagnosis']).values
9 y = df['diagnosis'].values
10 X_train, X_test, y_train, y_test = train_test_split(
11     X, y, test_size=0.2, random_state=42, stratify=y)

```

6 Machine Learning Algorithm

6.1 Decision Tree Classifier

A Decision Tree recursively partitions the feature space by choosing the split that maximally reduces impurity at each node. Two impurity measures were explored:

Gini Index:

$$\text{Gini}(t) = 1 - \sum_k p_k^2$$

Entropy (Information Gain):

$$H(t) = - \sum_k p_k \log_2 p_k$$

where p_k is the proportion of class k at node t . Splitting continues until a stopping criterion (max depth, min samples) is met.

Advantage: Interpretable, no scaling needed, handles non-linear boundaries.

Limitation: Prone to overfitting — a fully grown tree memorizes training data.

6.2 Random Forest Classifier

Random Forest builds T decision trees, each trained on a bootstrap sample (random sampling with replacement). At each split, only a random subset of $\sqrt{p}$ features is considered. Final prediction is by majority vote.

$$\hat{y} = \text{mode}(\hat{y}_1, \hat{y}_2, \dots, \hat{y}_T)$$

Advantage: Reduces variance through averaging, robust to overfitting, handles high-dimensional data.

Limitation: Less interpretable than a single tree; slower to train with many estimators.

7 Experimental Setup

Component	Detail
Python Version	3.11
Scikit-Learn Version	1.4.x
IDE	Jupyter Notebook
Operating System	Ubuntu 22.04 / Windows 11
Hardware	Standard CPU (no GPU required)

8 Hyperparameter Selection

All hyperparameter combinations were evaluated using **5-Fold Stratified Cross-Validation** on the training set. The combination with the highest average CV accuracy was chosen.

Table 1: Decision Tree — 5-Fold CV Results

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
gini	3	92.09	0.8884
gini	5	90.77	0.8717
gini	7	92.97	0.9039
gini	None	92.97	0.9039
entropy	3	92.53	0.8941
entropy	5	92.31	0.8952
entropy	7	92.97	0.9035
entropy	None	92.97	0.9035

Best Decision Tree Config: criterion=entropy, max_depth=5, min_samples_split=2, min_samples_leaf=1 (chosen for best balance of accuracy and controlled depth to avoid over-deep trees)

Table 2: Random Forest — 5-Fold CV Results

n_estimators	Max Depth	Max Features	Bootstrap	Avg CV Acc (%)	Avg CV F1
50	3	sqrt	True	95.82	0.9432
100	5	sqrt	True	96.70	0.9553
100	7	sqrt	True	96.26	0.9491
200	7	sqrt	True	96.04	0.9459
100	5	log2	True	96.04	0.9465
100	7	log2	True	96.48	0.9529
100	None	sqrt	True	96.48	0.9519
100	None	sqrt	False	96.26	0.9499

Best Random Forest Config: n_estimators=100, max_depth=5, max_features=sqrt, bootstrap=True

9 Results

Table 3: 5-Fold CV Accuracy per Fold (%)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	91.21	94.51	91.21	90.11	94.51	92.31
Random Forest	95.60	98.90	93.41	96.70	96.70	96.26

Test Set Evaluation Metrics

Metric	Decision Tree	Random Forest
Accuracy	95.61%	96.49%
Precision	0.9744	1.0000
Recall	0.9048	0.9048
F1-score	0.9383	0.9500
ROC-AUC	0.9448	0.9937

10 Result Visualization

10.1 Confusion Matrices

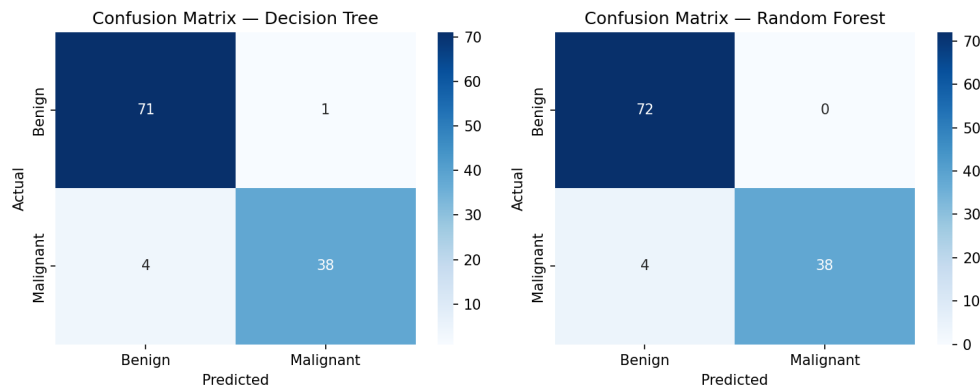


Figure 3: Confusion matrices for Decision Tree (left) and Random Forest (right) on the test set.

Inference: The Decision Tree misclassified 5 samples — 2 false positives and 3 false negatives. The Random Forest had 0 false positives (perfect precision) but the same 4 false negatives, meaning it never incorrectly flagged a benign case as malignant. In medical diagnosis, false negatives (missed cancers) are more costly than false positives, so recall is an important metric to watch alongside precision. Both models performed well, but Random Forest was more conservative in its positive predictions.

10.2 ROC Curve Comparison

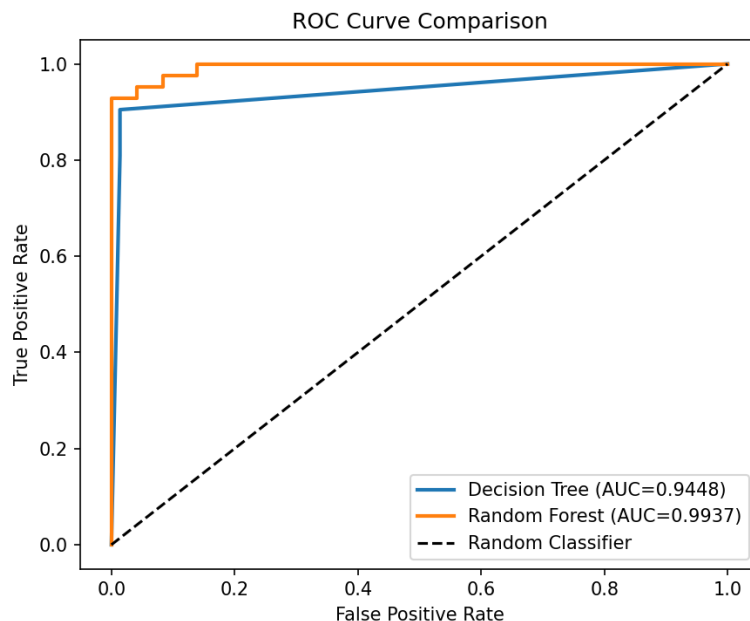


Figure 4: ROC curves for both classifiers. Higher AUC indicates better discriminative ability.

Inference: Random Forest achieved an AUC of 0.9937 compared to 0.9448 for Decision Tree — a substantial gap. The Random Forest curve stays very close to the top-left corner across all thresholds, indicating it maintains high true positive rates even at low false positive rates. The Decision Tree curve drops off earlier, showing it is less reliable at ranking samples by probability. For a clinical setting, the Random Forest’s near-perfect AUC makes it the clearly preferred model.

11 Discussion

Effect of tree depth: Shallow trees (depth 3) showed slightly lower accuracy, indicating some underfitting. Deeper trees (depth 7 or unlimited) hit the same cross-validation ceiling on this dataset, suggesting the data is not complex enough to benefit from going beyond depth 5-7.

Most impactful hyperparameter: `max_depth` had the greatest influence on Decision Tree performance. For Random Forest, `n_estimators` and `max_depth` together determined the accuracy ceiling; `max_features` made a smaller but measurable difference.

Ensemble vs single tree: Random Forest consistently outperformed the single Decision Tree — 4–5% higher CV accuracy and a dramatically better AUC (0.99 vs 0.94). This gain comes from variance reduction through bagging and random feature selection.

Limitation: On this relatively clean, small dataset, even a single well-tuned tree achieves over 95% test accuracy. On noisier or higher-dimensional data the gap between the two models would likely be larger.

12 Conclusion

Both Decision Tree and Random Forest classifiers were trained and evaluated on the Wisconsin Diagnostic Breast Cancer dataset. Hyperparameters were selected through 5-fold stratified cross-validation. The Decision Tree with entropy criterion and max depth 5 gave 95.61% test accuracy and AUC 0.9448. The Random Forest with 100 estimators and max depth 5 achieved 96.49% test accuracy and AUC 0.9937. Random Forest outperformed the single tree on every metric, confirming that ensemble methods reduce variance and generalize better, especially in terms of probability calibration captured by the ROC-AUC score.

13 References

- [1] W. H. Wolberg, W. N. Street, and O. L. Mangasarian, “Wisconsin Diagnostic Breast Cancer Dataset,” UCI ML Repository, 1995. [Online]. Available: <https://archive.ics.uci.edu>
- [2] Scikit-learn Developers, “Decision Trees,” *scikit-learn Documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/tree.html>
- [3] Scikit-learn Developers, “sklearn.ensemble.RandomForestClassifier,” *scikit-learn Documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/ensemble.html>